# e.mappa

## Deployment Readiness, Stakeholder Onboarding & Transparent Energy Asset Flows

*Working System Design Document | April 2026*

**Core conclusion**
e.mappa should not deploy solar first and hunt for demand later. It should qualify buildings, pre-onboard residents, secure deal-level capital, verify installation quality, and only then activate settlement. The flywheel is: high-readiness buildings -> high utilization -> credible payouts -> visible proof -> easier funding -> faster deployment.

## 1. Executive Summary

This document consolidates the operational conclusions from the current e.mappa strategy conversation. It defines the Deployment Readiness Score (DRS), the transparent deal-level funding model, fractional ownership mechanics, and detailed stakeholder onboarding/UI flows for residents, building owners, providers, financiers, electricians/labor network, suppliers, and the e.mappa internal operating layer.

- **Non-negotiable doctrine:** prepaid-only. No prepaid cash, no solar allocation, no stakeholder payout.
- **Economic base:** monetized solar, not generated solar. Unused, wasted, curtailed, or free-exported energy should not trigger payouts.
- **Deployment gate:** a building must pass DRS before capital is released and before the system goes live.
- **Funding model:** no opaque financier pool at the beginning. Use deal-level funding with full project transparency.
- **Transparency moat:** each stakeholder sees accurate counterpart performance benchmarks without exposing private financial details.
- **Ownership rule:** cashflows follow ownership. If a resident buys part of a panel array, that share of future energy cashflow moves with the ownership share.
- **Network scope:** the deployment network includes certified electricians/labor/fabrication under electrician-led execution, and suppliers as their own marketplace role.

## 2. The Flywheel We Are Actually Building

| Step | What happens | Why it matters |
|---|---|---|
| 1. Qualify building | DRS checks demand, prepaid commitments, load profile, installation readiness, installer readiness, supplier availability, and capital alignment. | Prevents bad deployments before money is spent. |
| 2. Pre-onboard residents | Residents register and preload solar tokens before go-live. | Turns demand from a hope into a measurable commitment. |
| 3. Fund specific deal | Financiers commit to a named building with disclosed assumptions and risks. | Creates trust and eliminates opaque capital allocation. |
| 4. Deploy through certified network | Electricians and labor/fabricators | Makes deployment repeatable and |

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| | follow verified checklists; suppliers fulfill standardized bills of materials. | auditable. |
| 5. Activate only after verification | Meter placement, inverter data, connectivity, and settlement rules are confirmed. | Protects safety, payout integrity, and reputation. |
| 6. Publish live performance | Stakeholders see role-specific performance dashboards and anonymized market benchmarks. | The second building becomes easier than the first because proof replaces pitch. |

# 3. Deployment Readiness Score (DRS)

DRS is the operating gate that decides whether a building is allowed to receive funding, hardware, installation scheduling, and go-live activation. It is not a marketing score. It is a risk-control mechanism designed to prevent underutilized, poorly verified deployments.

**DRS rule**

If a project does not pass DRS, it should not deploy. The cost of a blocked deployment is frustration. The cost of a bad deployment is broken trust across every stakeholder group.

| Score Band | Decision | Action |
|---|---|---|
| 80-100 | Auto-approve | Project can proceed to final funding, supplier lock, installer scheduling, and resident go-live prep. |
| 65-79 | Manual review | Ops reviews weak components. Project may proceed only after specific remediation. |
| Below 65 | Block | No deployment. Improve demand, prepayment, load profile, site readiness, or capital terms first. |

## 3.1 DRS Formula

| Component | Weight | What it measures | Hard notes |
|---|---|---|---|
| Demand Coverage | 35% | Projected monetized solar demand relative to expected solar output. | If projected utilization is below 60%, auto-block. |
| Prepaid Commitment | 20% | Real resident cash committed before activation, not just interest or signups. | No prepaid funds means no deployment. |
| Load Profile Quality | 15% | Whether demand happens when solar is produced or can be shifted through battery. | Daytime loads are especially valuable. |
| Installation Readiness | 10% | DB capacity, meter arrangement, cable route, rooftop, connectivity, permissions. | Unknowns are penalized heavily. |
| Installer/Labor Readiness | 10% | Certified electrician assigned, checklist readiness, safety standards. | No certified lead electrician means block. |
| Capital Alignment | 10% | Funding is secured and projected payback is within acceptable range. | If returns only work at unrealistic utilization, penalize. |

DRS = 0.35(Demand) + 0.20(Prepaid) + 0.15(Load Profile) + 0.10(Installation) + 0.10(Installer/Labor) + 0.10(Capital)

### 3.2 DRS Kill Switches

- Projected utilization below 60%: block.
- No prepaid funds: block.
- No certified lead electrician: block.
- Meter/inverter mismatch unresolved: block.
- Building owner permissions incomplete: block.
- No verified supplier quote/BOM for critical components: block.
- Connectivity for monitoring unavailable or unresolved: block.
- Settlement data cannot be trusted: block go-live even if installation is physically complete.

## 4. Transparent Deal-Level Funding, Not an Opaque Pool

The financier pool should not be used in the early system. It may become viable later, but only after enough trusted performance data exists. Early-stage capital should be radically transparent and tied to specific buildings.

| Funding Choice | Pros | Cons | Recommendation |
|---|---|---|---|
| Opaque pooled fund | Fast capital deployment once trust exists. | Too abstract early; financiers cannot see exact risk. | Do not use initially. |
| Building-specific funding | Maximum transparency; each financier knows the exact building, assumptions, risk, and return path. | Slower; more deal-by-deal friction. | Use for pilot and early scale. |
| Curated semi-pool | Financiers approve categories or risk bands; platform allocates within constraints. | Requires validated data and strong trust. | Possible Phase 2. |
| Algorithmic pool | Fastest and most scalable. | Requires deep historical performance and strong governance. | Only after the system earns credibility. |

### 4.1 What a Financier Must See Before Committing

- Exact building project, not a hidden allocation.
- DRS score and component breakdown.
- Total capital required and funding progress.
- Projected utilization assumptions and downside cases.
- Expected payback range, not a guaranteed return.
- Payout waterfall for recovery phase and royalty phase.
- Known risks: installation, demand, regulatory/metering, downtime, supplier delays.
- Live data after activation: utilization, revenue, payout, downtime, and variance from projection.

## 5. Stakeholder Value Proposition: Current Situation -> With e.mappa

| Stakeholder | Today | With e.mappa | Core pitch |
|---|---|---|---|
| Building Owner | Pays for electrical infrastructure but usually earns nothing from electricity. Electricity is a | Hosts a verified energy economy, earns hosting royalties, improves tenant appeal, and turns the | Your building stops being passive wiring and becomes a productive energy site. |

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| | cost center and tenant complaint source. | building into an energy-enabled asset. | |
| Resident | Pays utility directly, has no ownership, little transparency, and no upside. | Buys cheaper prepaid solar when available, falls back to grid, and can optionally own shares in local energy assets. | From electricity consumer to participant-owner. |
| Solar Provider | Must find customers, manage installations, and carry demand risk alone. | Deploys panels into qualified buildings, earns from monetized energy, and may sell ownership shares to recover capital while retaining upside on unsold shares. | Panels become liquid, yield-generating assets. |
| Infrastructure Financier | Infrastructure investing is high-ticket, opaque, and hard to monitor. | Funds specific building infrastructure with transparent assumptions, live performance tracking, priority recovery, and possible royalty phase. | Transparent granular infrastructure investing. |
| Electrician / Labor / Fabricator | One-off jobs with weak recurring pipeline and limited reputation compounding. | Certified network with repeat installation, maintenance, inspection, and fabrication workflows. | From job-by-job labor to trusted deployment network. |
| Supplier | Sells hardware through fragmented relationships and manual quoting. | Receives standardized BOM requests, competes transparently, fulfills verified project orders, and builds performance history. | From informal sales to trusted energy supply chain node. |
| e.mappa | Could be perceived as another solar marketplace if poorly positioned. | Becomes the coordination, settlement, transparency, and deployment operating system for building-level energy economies. | Not selling solar; operating the energy economy. |

# 6. Fractional Ownership and Share Mechanics

**Do not double count**

A provider or financier cannot sell ownership and still keep 100% of the cashflow from the sold portion. Cashflows must follow ownership. The powerful version is: recover capital early by selling shares while continuing to earn from the portion retained.

## 6.1 Provider Share Sale Model

A provider may install panels and initially own 100% of the provider array. Residents can buy fractional ownership of that array. When ownership transfers, future energy payouts transfer proportionally.

| Event | Example | Cashflow result |
|---|---|---|
| Provider owns 100% | Provider owns all panels on MPPT/provider allocation. | Provider receives 100% of provider-side payout attributed to that array. |
| Resident buys 30% | Resident pays for 30% economic ownership of the provider array. | Resident receives 30% of future provider-side payout; provider retains 70%. |
| Provider sells another 20% | Residents now own 50%; provider owns 50%. | Provider has received liquidity/capital recovery, but ongoing payout is now 50%. |

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|---|---|---|
| Panel produces less | Energy payout falls for all owners proportionally. | No guaranteed payout from unavailable or unmonetized energy. |

### 6.2 Financier Buyout / Infrastructure Share Model

The same principle applies to infrastructure financing. If residents or another buyer purchase an economic share of the financed infrastructure, the financier receives buyout cash and the future financier cashflow reduces proportionally. If no buyout happens, financiers continue according to the recovery and royalty terms.

- Recovery phase: financiers receive priority payout until principal plus target return is reached.
- Royalty phase: financiers step down to a smaller lifetime royalty tied only to monetized prepaid solar.
- Buyout/share transfer: cash paid now reduces future entitlement in exact proportion to ownership transferred.
- No hidden debt: if energy is not monetized, there is no payout obligation.

### 6.3 Ownership Ledger Requirements

- Every asset has an asset ID: provider array, inverter, battery, electrical balance-of-system, supplier order, building project.
- Every owner has an ownership percentage by asset or revenue stream.
- Every settlement period calculates payouts from monetized energy and current ownership percentages.
- Transfers update future cashflows only after payment clears and compliance checks pass.
- Audit trail is immutable: who owned what, when, at what price, and what payout rights moved.

## 7. Role-Specific Transparency Layer

The platform should show counterpart performance, but in a privacy-safe way. The rule is: show distributions, benchmarks, and verified outcomes; do not expose private identities, addresses, raw income, or personal financial details unless the stakeholder explicitly owns or funds that project.

| Role | Can see about themselves | Can see about counterparts | Should not see |
|---|---|---|---|
| Building Owner | Own building utilization, host royalty, DRS, maintenance status. | Anonymized comparable building earnings and utilization bands. | Exact income/address of another owner. |
| Provider | Own array output, payout, retained ownership, sold shares, payback progress. | Provider ROI/payback bands by building type/utilization. | Another provider's raw earnings unless anonymized/aggregated. |
| Financier | Own funded project performance, payback curve, payout history, variance. | System-wide on-track rates, utilization distributions, risk bands. | Unrelated financiers' exact balances or private deal terms. |
| Resident | Own savings, solar usage, prepaid balance, owned shares. | Building-level utilization and optional anonymized owner/share benchmark. | Other residents' balances, consumption, or earnings. |
| Electrician/Labor | Own jobs, checklist performance, quality score, payout status. | Benchmark install success rates and response times. | Sensitive project financing details. |
| Supplier | Own quotes, orders, fulfillment score, payment status. | Anonymized supplier performance benchmarks. | Competitor exact margins or private cost structures. |

# 8. Detailed Onboarding and UI Flows

The following flows describe what each stakeholder sees and clicks. The guiding principle: each role should see enough truth to make a confident decision, but not so much raw complexity that they freeze.

## 8.1 Building Owner Flow

Building owners need to understand whether their property qualifies, what they earn, what disruption to expect, and how their building compares to others.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Welcome / Role Selection | CTA: “List my building.” Copy: “Turn unused rooftop and electrical access into a verified local energy economy.” |
| 2 | Screen 2: Building Basics | Inputs: location, number of units, occupancy, building type, roof type, meter type if known, contact person. Output: preliminary eligibility. |
| 3 | Screen 3: Current Situation vs e.mappa | Shows: “Today: you provide electricity access but usually earn nothing from utility usage. With e.mappa: your building can earn hosting royalties and attract residents with cheaper prepaid solar.” |
| 4 | Screen 4: Comparable Buildings | Shows anonymized benchmarks: host earnings range, utilization range, solar coverage, number of similar buildings. No addresses or private finances. |
| 5 | Screen 5: Site Inspection Request | Owner chooses inspection slot or submits photos: DB, meters, roof, cable route, connectivity, permission confirmation. |
| 6 | Screen 6: DRS Dashboard | Displays DRS total and component scores. Weak areas are actionable: “Need 10 more resident pre-registrations” or “Meter placement unclear.” |
| 7 | Screen 7: Resident Pre-Onboarding Toolkit | Owner receives invite link/QR, tenant message template, poster, and progress tracker: registered units, prepaid units, projected utilization. |
| 8 | Screen 8: Deal Approval | Owner reviews host royalty terms, access obligations, maintenance access, minimum term, dispute rules, and agrees digitally. |
| 9 | Screen 9: Deployment Tracker | Shows funding progress, supplier status, assigned electrician, installation date, verification status. |
| 10 | Screen 10: Live Dashboard | After go-live: monthly host royalty, building utilization, solar coverage, uptime, resident adoption, maintenance events, comparable benchmark movement. |

## 8.2 Resident Flow

Residents need cheap, simple, reliable energy first. Ownership should be optional and introduced only after they trust the basic utility experience.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Join Building | Resident enters building code/QR and phone number. OTP verification. |
| 2 | Screen 2: What Changes for You | Copy: “Use cheaper prepaid solar when available. If solar runs out, you fall back to grid. No postpaid bills.” |
| 3 | Screen 3: Preload Solar Tokens | Resident tops up before go-live. Show launch bonus only if economics allow; never promise unlimited solar. |
| 4 | Screen 4: Solar Availability Explanation | Simple graphic: solar first, battery second, grid fallback third. Explain that balance only buys e.mappa solar, not KPLC tokens. |
| 5 | Screen 5: Home Dashboard | Cards: solar balance, today’s solar used, savings vs grid, grid fallback, building utilization. |
| 6 | Screen 6: Optional Ownership Intro | After usage history: “Own a share of local solar assets.” Show risk: earnings depend on monetized solar, not sunlight alone. |
| 7 | Screen 7: Share Marketplace | Resident can buy available provider shares or infrastructure shares. Shows asset, price, expected payout range, utilization history, risk notes. |
| 8 | Screen 8: Ownership Dashboard | Shows owned percentage, energy payout earned, payback progress, transfer/sell option if enabled, and audited settlement statement. |
| 9 | Screen 9: Alerts | Low balance, solar unavailable, maintenance outage, unusually high grid fallback, share payout received. |

## 8.3 Provider Flow

Providers need to see that panels are not just hardware: they are deployable, trackable, partially liquid, yield-generating assets.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Provider Signup | Inputs: name/entity, location, hardware available, panel specs, proof of ownership/purchase, payout account. |
| 2 | Screen 2: Current Situation vs e.mappa | Copy: “Today you must find customers and carry demand risk. With e.mappa you deploy panels into DRS-qualified buildings and earn from monetized energy.” |
| 3 | Screen 3: Available Projects | Shows buildings seeking provider capacity, DRS, projected utilization, rooftop capacity, expected provider payback range, required panel specs. |
| 4 | Screen 4: Commit Panels | Provider chooses project and capacity. App checks compatibility: wattage, voltage/stringing, MPPT constraints, physical mounting needs. |
| 5 | Screen 5: Terms and Risk | Provider confirms payout is based on energy sold, not energy generated; panel failure reduces payout; ownership transfer reduces future payout. |

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| 6 | Screen 6: Share Sale Option | Provider chooses whether to sell 0-100% of future provider-side ownership. Shows tradeoff: upfront liquidity vs retained yield. |
| 7 | Screen 7: Installation Coordination | Assigned electrician/labor schedule, supplier dependencies, mounting/fabrication needs, inspection checklist status. |
| 8 | Screen 8: Live Asset Dashboard | Cards: kWh generated, kWh monetized, utilization, payout, retained ownership, sold ownership, payback progress. |
| 9 | Screen 9: Provider Market Benchmarks | Shows anonymized ROI/payback bands by utilization, building type, and panel size. No other provider's exact private data. |
| 10 | Screen 10: Transfer/Sell More Shares | Provider can list additional ownership shares if allowed. App previews future cashflow reduction before listing. |

## 8.4 Financier Flow

Financiers need radical transparency: a named project, exact assumptions, downside cases, live monitoring, and no hidden pooling.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Financier Signup | KYC/entity details, payout account, risk preference, ticket size. |
| 2 | Screen 2: Current Situation vs e.mappa | Copy: "Traditional infrastructure investing is opaque and high-ticket. e.mappa lets you fund specific building energy assets with live performance data." |
| 3 | Screen 3: Deal Marketplace | Each card: building nickname/code, city/area band, DRS, capital needed, funded percentage, projected utilization, expected recovery range. |
| 4 | Screen 4: Deal Room | Detailed project: hardware BOM, supplier quotes, installer, site inspection, owner terms, resident prepayment, risks, payout waterfall. |
| 5 | Screen 5: Stress Test View | Shows base, downside, severe downside: utilization, revenue, payback period, royalty phase estimate. No guaranteed returns. |
| 6 | Screen 6: Commit Capital | Financier selects amount. Funds are escrowed/released by milestones: inspection passed, supplier order, install complete, verification complete. |
| 7 | Screen 7: Activation Gate | Financier sees whether project passed final verification. If not, capital release pauses under predefined rules. |
| 8 | Screen 8: Live Payback Dashboard | Projected vs actual payback curve, cumulative payout, utilization, downtime, variance from plan, next expected settlement. |
| 9 | Screen 9: Transparency Benchmarks | System-wide anonymized data: % buildings above utilization bands, median recovery, delayed projects, maintenance frequency. |

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|---|---|---|
| 10 | Screen 10: Exit/Transfer Option | If enabled: financier can sell remaining economic rights to another financier/resident. Future payout entitlement transfers after settlement. |

## 8.5 Electrician / Labor / Fabricator Network Flow

Electricians are the certified execution lead. Fabricators and general labor sit under or are coordinated by the electrician workflow so responsibility stays clear.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Electrician Signup | License/experience, service area, references, specialties, phone, payout account. |
| 2 | Screen 2: Certification Module | Training: e.mappa architecture, safety, meter placement documentation, DB connection, anti-islanding/zero-export requirements, data verification. |
| 3 | Screen 3: Practical Checklist Test | Must pass photo-based and scenario-based assessment. Certification is earned, not automatic. |
| 4 | Screen 4: Job Marketplace | Jobs: site inspection, installation, maintenance, verification, cable routing, DB modification, fabricator coordination. |
| 5 | Screen 5: Site Inspection Workflow | Photo requirements: DB, meter arrangement, cable route, roof, connectivity, building permission. App rejects incomplete inspection. |
| 6 | Screen 6: Installation Checklist | Step-by-step tasks: mounting, cable routing, protection devices, inverter, battery, monitoring dongle/router, labels, grounding, DB breaker, test readings. |
| 7 | Screen 7: Labor/Fabricator Subtasks | Electrician can request/assign fabrication/labor tasks: brackets/rails, conduit routing, mounting support, roof works. Electrician remains accountable. |
| 8 | Screen 8: Go-Live Verification | System cannot go live until readings, connectivity, safety photos, and e.mappa ops signoff are complete. |
| 9 | Screen 9: Performance Dashboard | Completion rate, failed inspections, uptime after install, response time, customer rating, certification tier. |
| 10 | Screen 10: Maintenance Tickets | Fault alerts, site access instructions, parts needed, repair verification, reserve-funded payout status. |

## 8.6 Supplier Flow

Suppliers need their own interface because hardware availability, quote accuracy, and fulfillment speed directly affect deployment reliability.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Supplier Signup | Business details, locations, product categories, brands carried, delivery regions, payment terms, contact |

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| | | person. |
| 2 | Screen 2: Product Catalog | Supplier uploads inventory: inverters, batteries, panels, SPDs, MCBs, cables, conduit, routers, mounting materials. Include specs, price, warranty, lead time. |
| 3 | Screen 3: Current Situation vs e.mappa | Copy: “Today hardware sales are relationship-driven and manual. With e.mappa you receive standardized project BOM requests and build verified fulfillment history.” |
| 4 | Screen 4: BOM Request Inbox | Project-specific RFQs with required specs, quantity, delivery location, deadline, substitution rules. |
| 5 | Screen 5: Quote Submission | Supplier submits price, availability, warranty, delivery timeline, substitution options, validity period. |
| 6 | Screen 6: Award / Purchase Order | If selected, supplier receives order, milestone terms, delivery instructions, and required documentation. |
| 7 | Screen 7: Fulfillment Tracker | Packing, dispatch, delivery confirmation, recipient signature/photo, serial numbers, warranty documents. |
| 8 | Screen 8: Supplier Performance Dashboard | Quote response time, price competitiveness, delivery reliability, defect/return rate, warranty responsiveness. |
| 9 | Screen 9: Market Benchmark | Anonymized: common components, demand by region, average lead times, price bands. No competitor margins. |
| 10 | Screen 10: Dispute/Warranty Workflow | Defective component claim, evidence upload, replacement approval, service-level tracking. |

## 8.7 e.mappa Internal Ops / AI Governance Flow

The internal layer watches the whole system: deployment risk, settlement integrity, utilization, payouts, anomalies, and stakeholder health.

| UI Step | Screen / Action | Required precision |
|---|---|---|
| 1 | Screen 1: Pipeline Dashboard | Buildings by stage: lead, inspection, resident pre-onboarding, funding, supplier, install, verification, live. |
| 2 | Screen 2: DRS Command Center | Score distribution, blocked projects, manual review queue, weak components, recommended remediation. |
| 3 | Screen 3: Settlement Monitor | E_gen, E_sold, E_waste, grid fallback, prepaid cash, payout waterfall, reserve balance, anomalies. |
| 4 | Screen 4: Counterparty Health | Provider ROI health, financier recovery health, owner royalty trends, resident savings, installer quality, supplier reliability. |
| 5 | Screen 5: Alert Inbox | Low utilization, missing inverter data, high waste, downtime, tamper suspicion, prepaid shortfall, supplier delay, installer verification failure. |

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|---|---|---|
| 6 | Screen 6: Governance Actions | Pause settlement, request inspection, block go-live, adjust DRS, flag deal risk, trigger maintenance reserve, recommend demand incentives. |
| 7 | Screen 7: Learning Loop | Compare DRS prediction vs actual utilization. Improve scoring weights over time, but never silently weaken risk controls. |
| 8 | Screen 8: Public Benchmark Engine | Generates privacy-safe aggregated benchmark cards for stakeholder interfaces. |

# 9. Exact Share Ownership UI Flow

## 9.1 Resident Buying Provider Shares

| Step | What resident sees/clicks | Backend action |
|---|---|---|
| 1 | Tap “Own Solar Shares.” | Fetch eligible assets in resident building and allowed marketplace listings. |
| 2 | Select provider array: shows installed capacity, utilization history, remaining shares, price per 1%, expected payout range. | Calculate current ownership ledger and historical monetized energy. |
| 3 | Risk disclosure: “Returns depend on prepaid solar sold, not generated solar. If utilization drops, payouts drop.” | Require acceptance and log timestamp. |
| 4 | Choose ownership amount, e.g., 5%. | Reserve listing temporarily and calculate price, fees, and future payout entitlement. |
| 5 | Pay via M-Pesa/card/wallet. | Hold funds until settlement and transfer conditions are satisfied. |
| 6 | Ownership confirmation. | Update ownership ledger: seller percentage down, resident percentage up. Future payouts update from next settlement period. |
| 7 | Dashboard updates. | Show owned %, payout earned, cumulative return, transfer option if enabled. |

## 9.2 Provider Selling Shares

| Step | What provider sees/clicks | Important safeguard |
|---|---|---|
| 1 | Tap “Sell Ownership Shares.” | Must show retained ownership and estimated future payout reduction. |
| 2 | Choose % to list and price. | Cannot list more than owned. Cannot hide risks. |
| 3 | Preview: upfront cash vs reduced future earnings. | Prevents misunderstanding of liquidity vs yield tradeoff. |
| 4 | Publish listing to eligible residents/investors. | Listing includes utilization history, not guaranteed returns. |
| 5 | Sale completed. | Provider receives cash after settlement; future provider payout reduces proportionally. |

### 9.3 Financier Infrastructure Buyout / Transfer

| Step | What happens | Result |
|---|---|---|
| 1 | Financier lists part of remaining economic rights or resident/investor buys out a defined share. | Listing shows project, DRS, actual performance, remaining recovery/royalty rights. |
| 2 | Buyer pays agreed price. | Funds held until transfer confirmation. |
| 3 | Ledger updates. | Seller future infrastructure payout decreases; buyer receives corresponding future entitlement. |
| 4 | Dashboard updates. | Both parties see updated ownership and expected payout curves. |

## 10. Deployment Sequence: The Correct Order

| Order | Stage | Gate | Output |
|---|---|---|---|
| 1 | Building lead capture | Owner submits basics. | Preliminary building profile. |
| 2 | Site inspection | Photos and checklist complete. | Installation readiness score. |
| 3 | Resident pre-onboarding | Enough residents register and preload tokens. | Demand and prepaid commitment score. |
| 4 | DRS calculation | Score >=80 or manual review clears weaknesses. | Deployment approval. |
| 5 | Deal-level funding | Named project receives committed capital. | Capital alignment. |
| 6 | Supplier RFQ/order | Verified BOM and supplier selected. | Hardware procurement. |
| 7 | Provider commitment | Panel providers commit compatible capacity. | Generation profile. |
| 8 | Certified installation | Electrician/labor execute checklist. | Physical system installed. |
| 9 | Verification | Meter/inverter/connectivity/ settlement verified. | Go-live approval. |
| 10 | Activation and settlement | Residents consume prepaid solar; payouts begin only on monetized kWh. | Live project data and stakeholder dashboards. |

## 11. Data Model: Minimum Objects Required

| Object | Fields | Why it matters |
|---|---|---|
| Building | ID, location band, units, occupancy, DRS, meter config, owner terms, live status. | Core deployment and benchmark unit. |
| Resident Account | Building ID, prepaid balance, consumption, ownership holdings, payout wallet. | Prepaid allocation and optional ownership. |
| Provider Asset | Provider ID, array capacity, MPPT/channel, ownership ledger, output, payout. | Panel-side generation and yield. |
| Infrastructure Asset | Inverter, battery, DB upgrades, cables, monitoring, financier ownership. | Capital recovery and maintenance accounting. |
| Supplier Order | BOM, supplier, quote, price, lead time, serials, warranty, delivery status. | Procurement reliability and auditability. |
| Installer Job | Electrician ID, checklist, photos, inspection result, go-live signoff. | Quality and safety enforcement. |

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|---|---|---|
| Settlement Period | E_gen, E_sold, E_waste, E_grid, prepaid cash, payouts, reserve. | Truth source for all dashboards. |
| Ownership Ledger | Asset ID, owner ID, percentage, effective date, transfer price, rights moved. | Prevents double-counting and cashflow disputes. |

## 12. Accuracy Rules and Anti-Fluff Guardrails

- Never promise guaranteed returns. Always show ranges tied to utilization and actual prepaid sales.
- Never pay stakeholders from generated-but-unsold energy.
- Never hide ownership dilution: share sales reduce future cashflow for the seller.
- Never deploy before demand is pre-committed and DRS clears.
- Never treat supplier availability as an afterthought; supplier delays can stall the flywheel.
- Never let electrician signup equal certification; quality must be earned and verified.
- Never show private counterpart finances; show anonymized benchmarks and distributions instead.
- Never let UI simplicity erase system truth; abstract complexity but preserve the underlying rules.

## 13. Final Product Positioning

**Positioning statement**

e.mappa turns apartment buildings into transparent, prepaid, locally financed energy economies where residents buy cheaper solar, providers deploy yield-generating panels, financiers fund verified infrastructure, suppliers and electricians power repeatable deployment, and building owners host productive energy assets.

### 13.1 The Sentence That Should Guide the Product

e.mappa does not win by installing solar. It wins by making sure every deployed kWh is pre-sold, verified, settled, and transparently attributed to the people who made it possible.

## Appendix A: Initial MVP Scope

| MVP Must Have | Can Wait |
|---|---|
| Building owner onboarding + site inspection + DRS | Fully algorithmic financier pool |
| Resident preregistration + prepaid token top-up | Advanced dynamic pricing |
| Deal-level financier funding room | Secondary marketplace for every asset type |
| Supplier RFQ/BOM workflow | Automated supplier auction optimization |
| Electrician certification + checklist + photo verification | Complex installer gamification |
| Provider asset dashboard + retained/sold share accounting | Public provider leaderboard |
| Settlement engine based on monetized prepaid kWh | AI autonomy over economic parameters |
| Privacy-safe benchmark cards | Full external public marketplace |

## Appendix B: Launch Metrics

| Metric | Target / Warning | Why |
|---|---|---|
| DRS at deployment | Target 80+; manual review 65-79; block <65 | Prevents bad deployments. |
| Projected utilization | Target 75%+; warning <60% | Primary economic health signal. |

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|---|---|---|
| Prepaid resident participation | Target enough cash-backed demand before go-live | Prevents wasted solar. |
| Supplier fulfillment reliability | High on-time delivery, low defect rate | Deployment repeatability. |
| Installer checklist pass rate | Must be very high before scaling | Safety and reputation. |
| Settlement data completeness | Near 100%; missing data pauses settlement | Trust and payout integrity. |
| Provider payback drift | Projected vs actual tracked monthly | Provider confidence. |
| Financier recovery drift | Projected vs actual tracked monthly | Capital confidence. |
| Resident savings | Positive vs grid price | Demand retention. |
| Owner royalty + occupancy signal | Tracked, but pitch should not rely only on small royalty | Owner adoption realism. |

## Source Notes

This document uses the current e.mappa working assumptions from the pilot hardware architecture, settlement/payback/stress-test blueprint, and project conversation conclusions. Specific figures such as pilot infrastructure cost, provider entry cost, prepaid-only doctrine, monetized-kWh settlement, and payback sensitivity should be revisited after live pilot data arrives.